

# ML FINAL EXAM

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# Project Overview & Methods

**Goal:** Build a classification model to predict whether an Airbnb listing is *instant\_bookable*.

**Dataset:** Airbnb Open Data (102,599 rows)

## Data Preparation:

- Standardized column names
- Cleaned price & service\_fee
- Imputed missing values (mode, median, “Unknown”)
- Removed zero/negative price rows
- Feature engineering:
  - fee\_to\_price\_ratio
  - construction\_year\_scaled
  - reviews\_log
- One-hot encoding for room\_type & cancellation\_policy
- Category codes for neighbourhood\_group & host\_identity\_verified
- Downsampled majority class
- Dropped non-predictive metadata
- Exported cleaned dataset to GitHub repo

**Train/Validation/Test Split:** 70% / 15% / 15%

# Models, Ensembles & Results

**Evaluation Metrics:** Accuracy, Precision, Recall, F1-Score, ROC-AUC

**Models Trained:**

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- KNN
- SVC

**Top Models:** Random Forest, Gradient Boosting, SVC

**Ensembles:**

- **Soft Voting Classifier** (top 3 models)
- **Bayesian Ensemble** (accuracy-weighted)

**Results:**

- Ensembles outperformed individual models
- Strong F1-Score and ROC-AUC
- Stable generalization on Test set

**Conclusion:** The project successfully predicts instant bookability using a full ML pipeline that meets all Final Exam requirements.